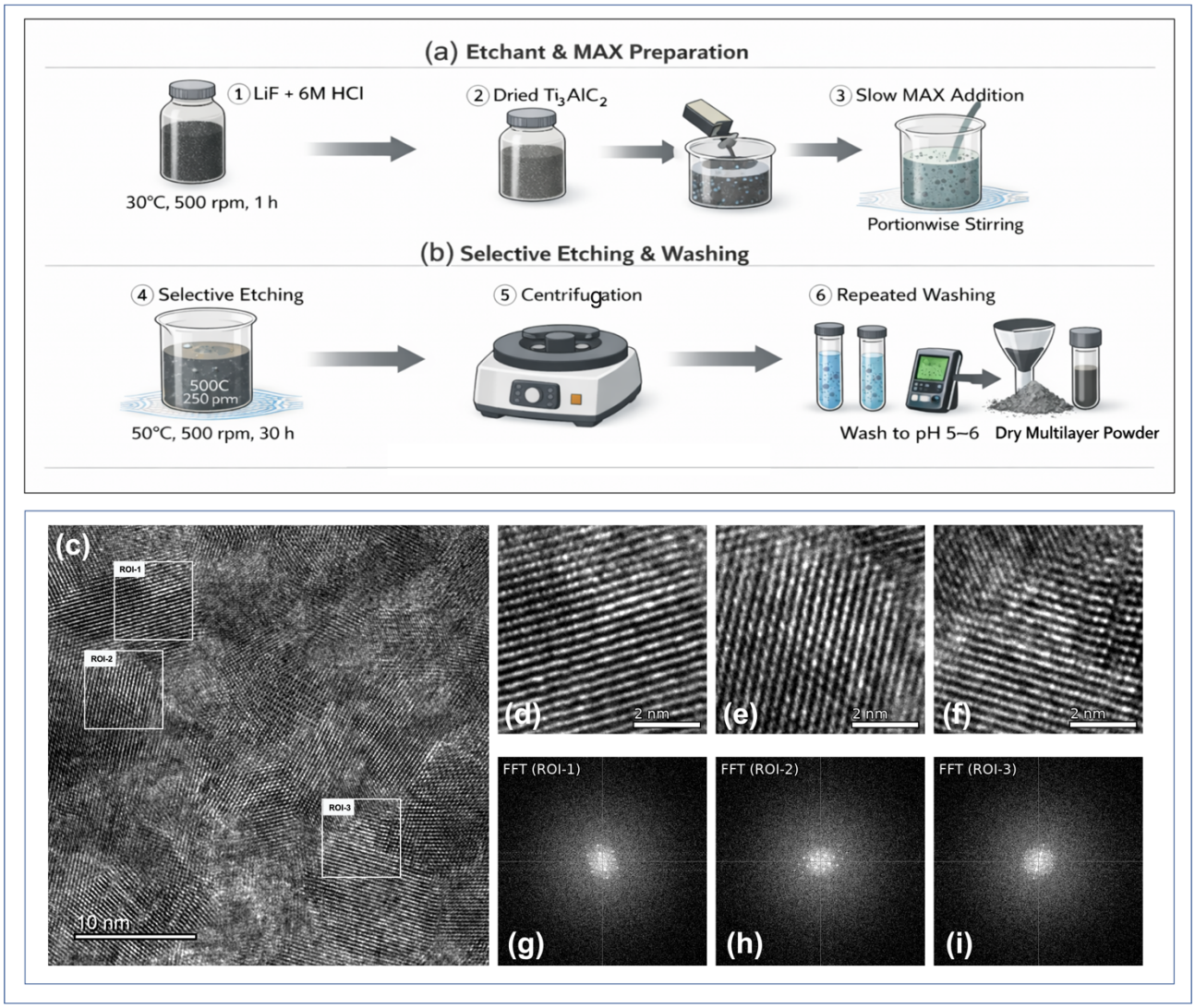Figure 8 | LiF/HCl synthesis workflow and lattice-resolved TEM validation of $Ti_3C_2T_x$ MXene. (a) Preparation of the LiF/6 M HCl etchant (30 °C, 500 rpm, 1 h) and MAX conditioning/portion wise Ti_3AlC_2 addition to ensure homogeneous wetting and reproducible etching. (b) Selective Al removal at 50 °C (500 rpm, 30 h), followed by separation and repeated washing/centrifugation to pH 5–6 to remove dissolved byproducts and residual acidity/fluorides while stabilizing surface terminations (T_x = –F/–OH/=O). (c) Representative HRTEM image (scale bar, 10 nm) revealing a mosaic of rotated crystalline domains consistent with few-layer flakes and partial restacking. (d)–(f) Lattice-fringe ROIs (scale bars, 2 nm). (g)–(i) Corresponding FFT patterns showing discrete reciprocal-space maxima. Dominant fringe spacings are **0.307, 0.300, and 0.294 nm** (ROIs 1–3), yielding **0.300 ± 0.007 nm** and an in-plane lattice constant $a = 0.347 \pm 0.008\, nm$ (indexed to $\{10\bar{1}0\}/\{100\}$).